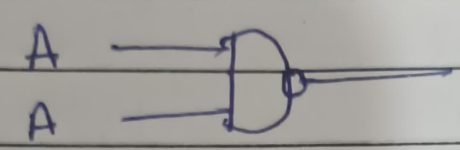
Page No.
Date

1. NAND se NOT.

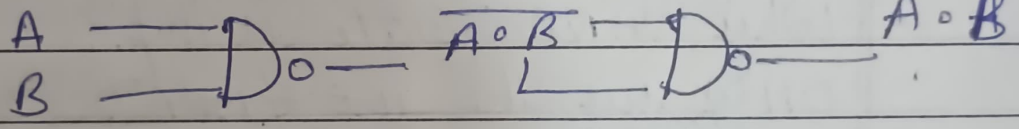
(do no input same hoga :). 1 input 1 NAND



Truth Table

A	NAND A	$A \cdot A$ NAND $\overline{A \cdot A} = \overline{A}$
0	1	
1	0	

2. NAND se ~~NOT~~ AND.



2 input
2 NAND

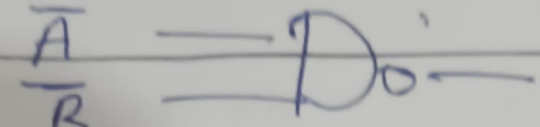
Truth table.

A	B	AND	NOT(AND) - NAND
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

Page No.
Date

(NOT me input)

3) NAND se OR. (NAND)



$\overline{A} \cdot \overline{B} = \overline{A} + \overline{B} = A + B$

Truth table

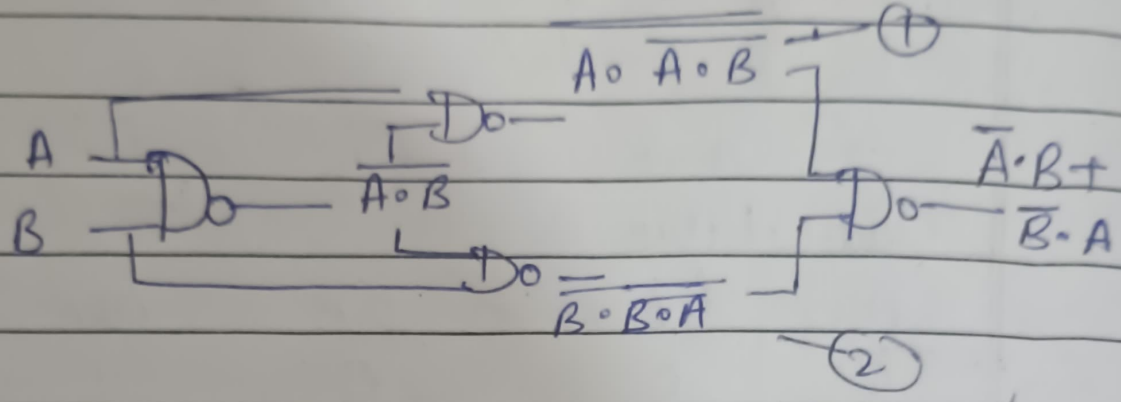
A	B	OR	NOR
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

4) NAND se XOR.

XOR me diff. input → output = 1
same input → output = 0

truth table

A	B	output
0	0	0
0	1	1
1	0	1
1	1	0



eq (1) $\overline{(A \cdot (\overline{A \cdot B}))}$	eq (2) $\overline{B \cdot (\overline{B \cdot A})}$
$\overline{A \cdot \overline{A} + A \cdot \overline{B}}$	$\overline{B \cdot \overline{B} + B \cdot \overline{A}}$
$\overline{0}$	$\overline{B \cdot \overline{A}}$
$\overline{A \cdot \overline{B}}$	$\overline{\overline{B} + \overline{\overline{A}}} = \overline{B} + A$
$= \overline{\overline{A}} + \overline{\overline{B}} = \overline{A} + B$	

$$\overline{(\overline{A} + B) \cdot (\overline{B} + A)}$$

$$\overline{A} \cdot \overline{B} + \overline{A} \cdot A + \overline{B} B + B \cdot A$$

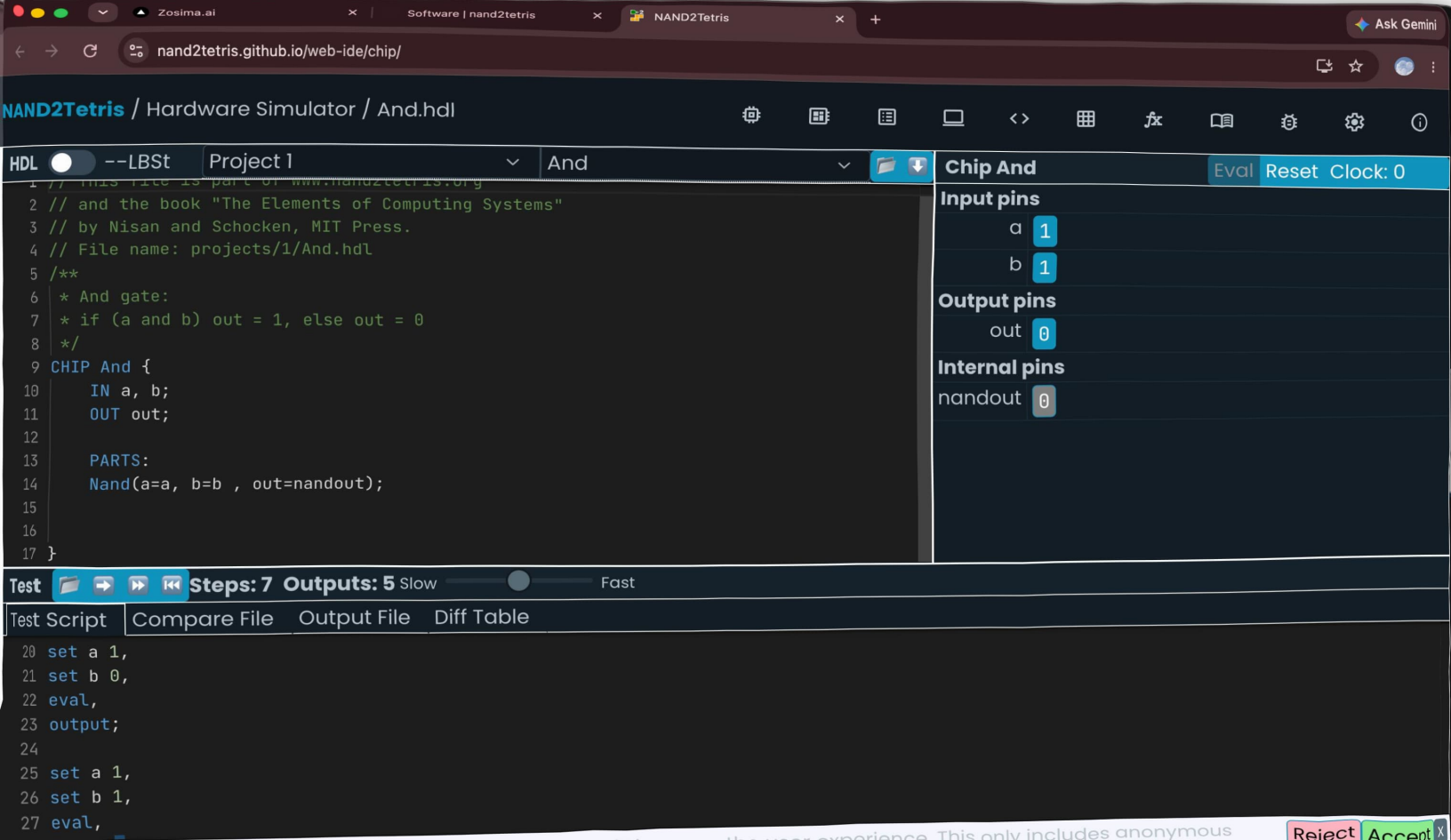
$$\overline{A} \cdot \overline{B} + B \cdot A$$

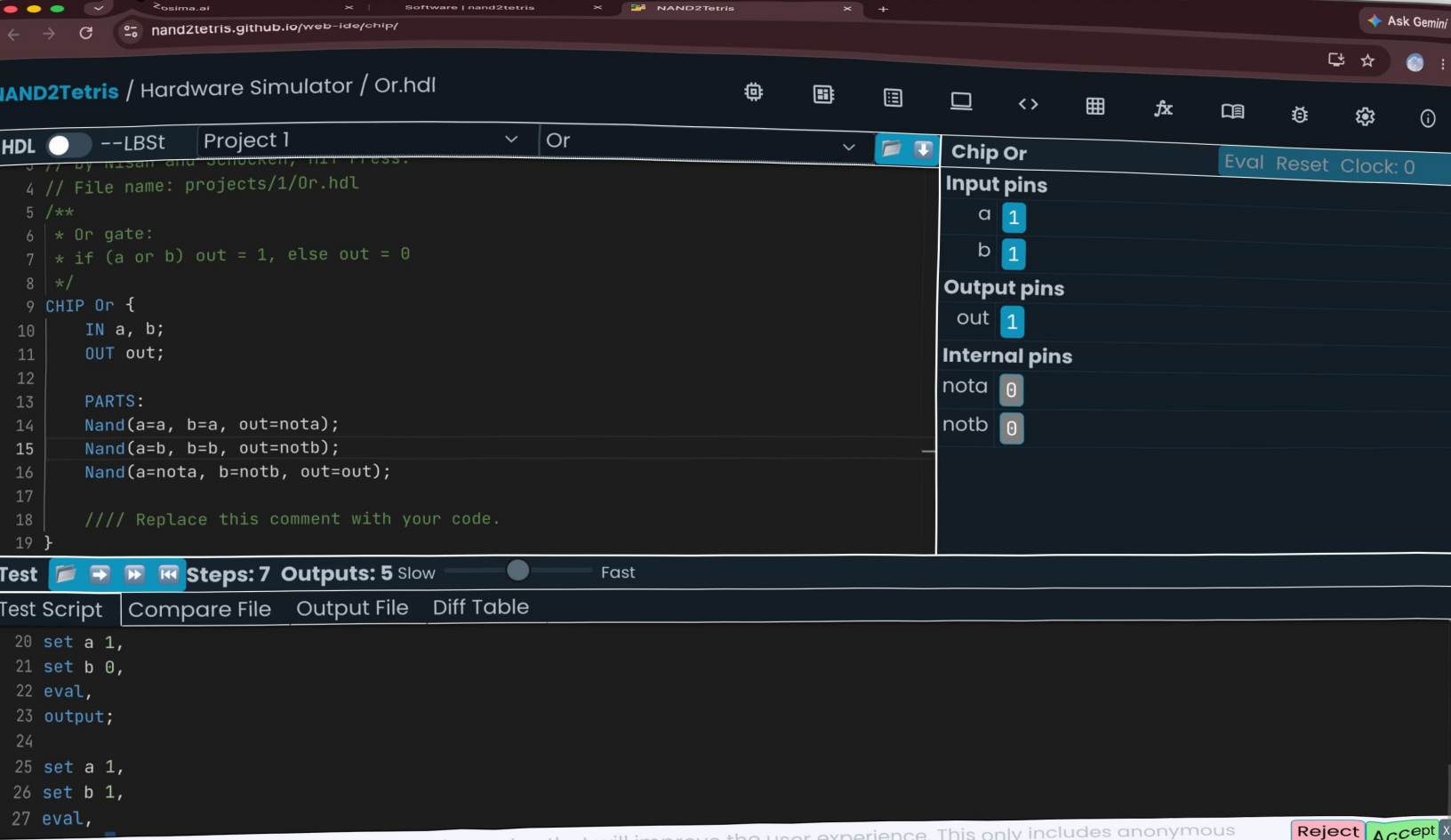
$$\overline{\overline{A}} + \overline{\overline{B}} + \overline{B} + \overline{A}$$

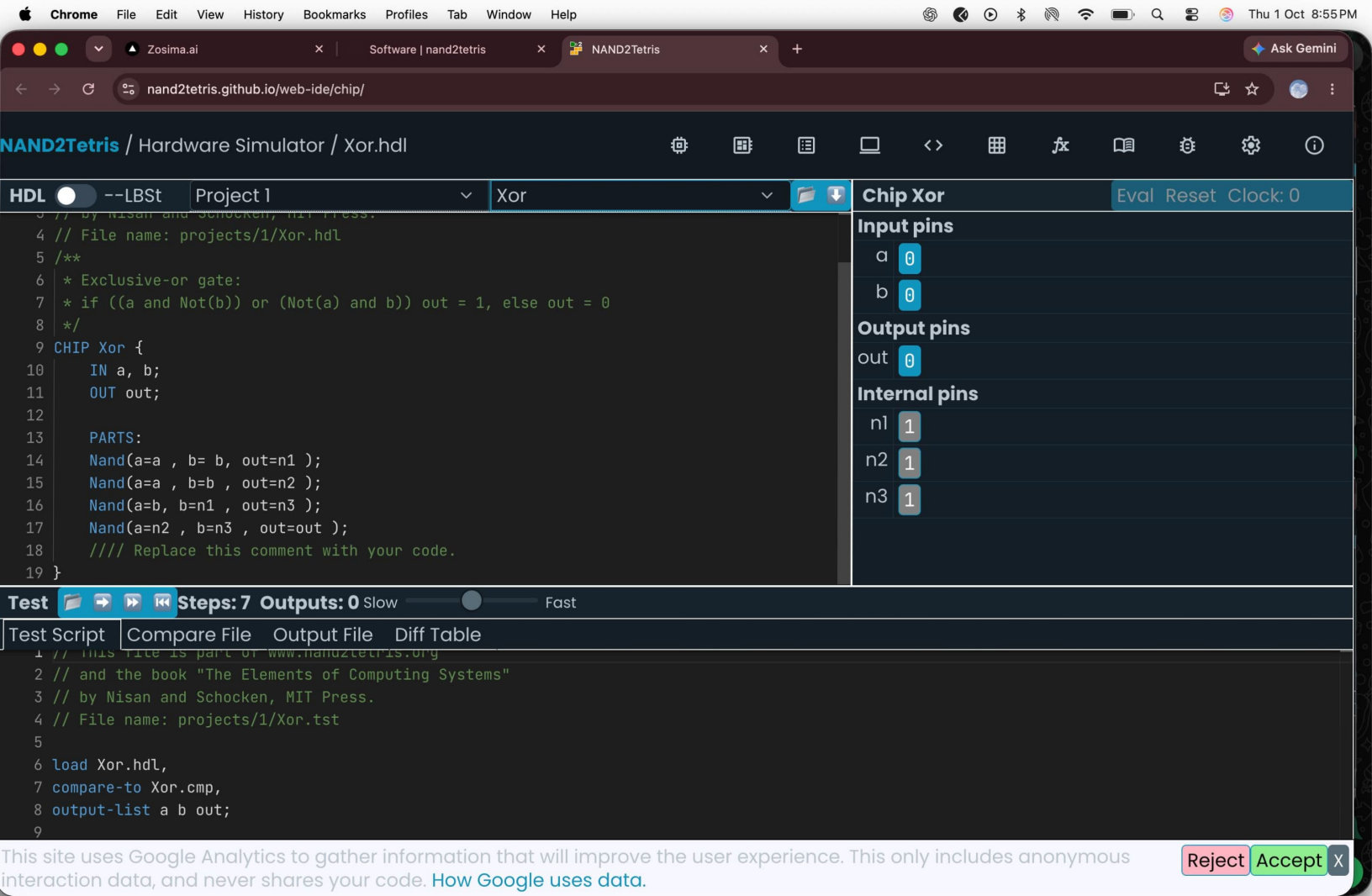
$$(\overline{A} + B) + (\overline{B} + \overline{A})$$

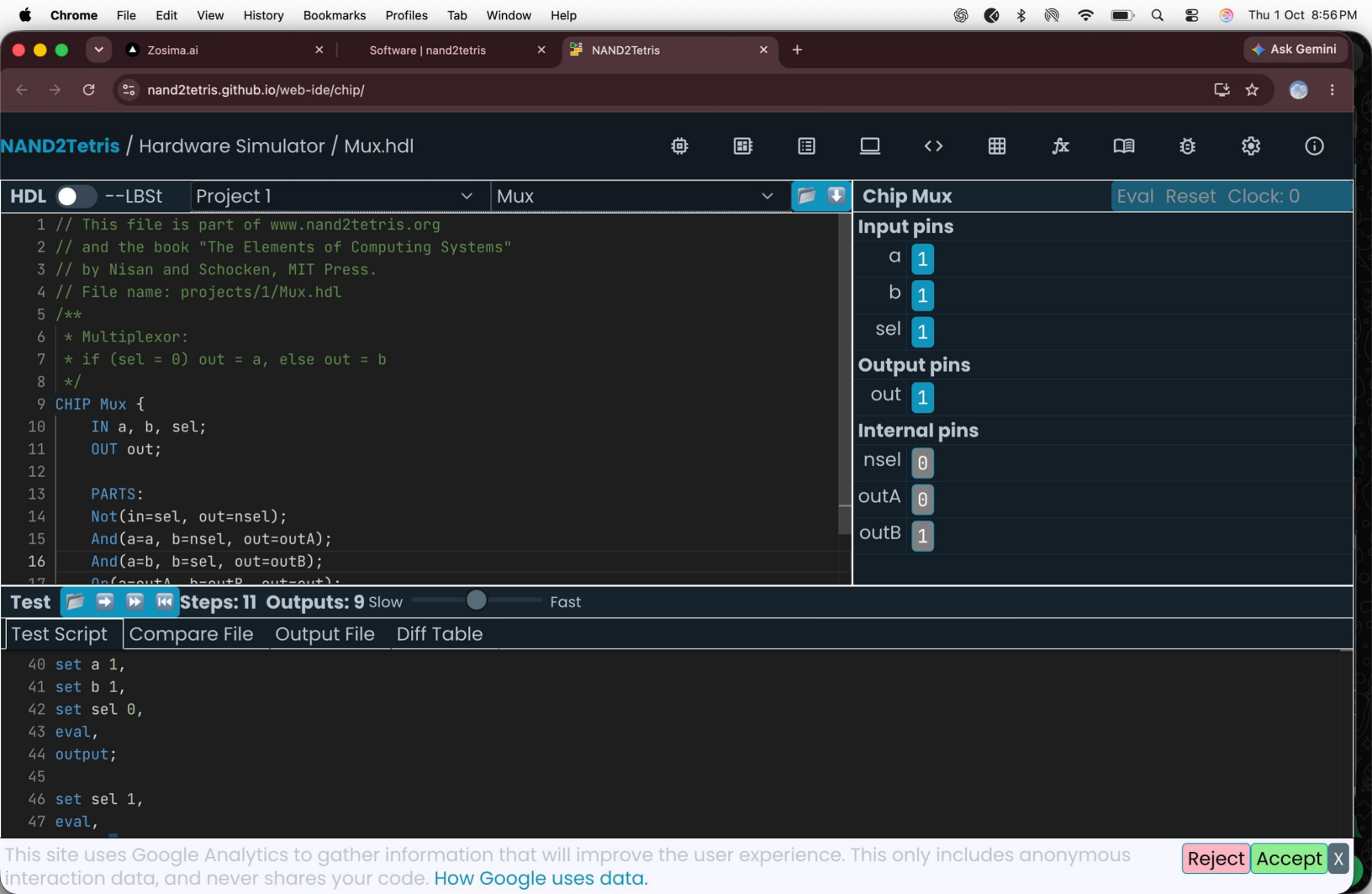
$$A \cdot \overline{B} + A \cdot \overline{A} + B + \overline{B} + B \overline{A}$$

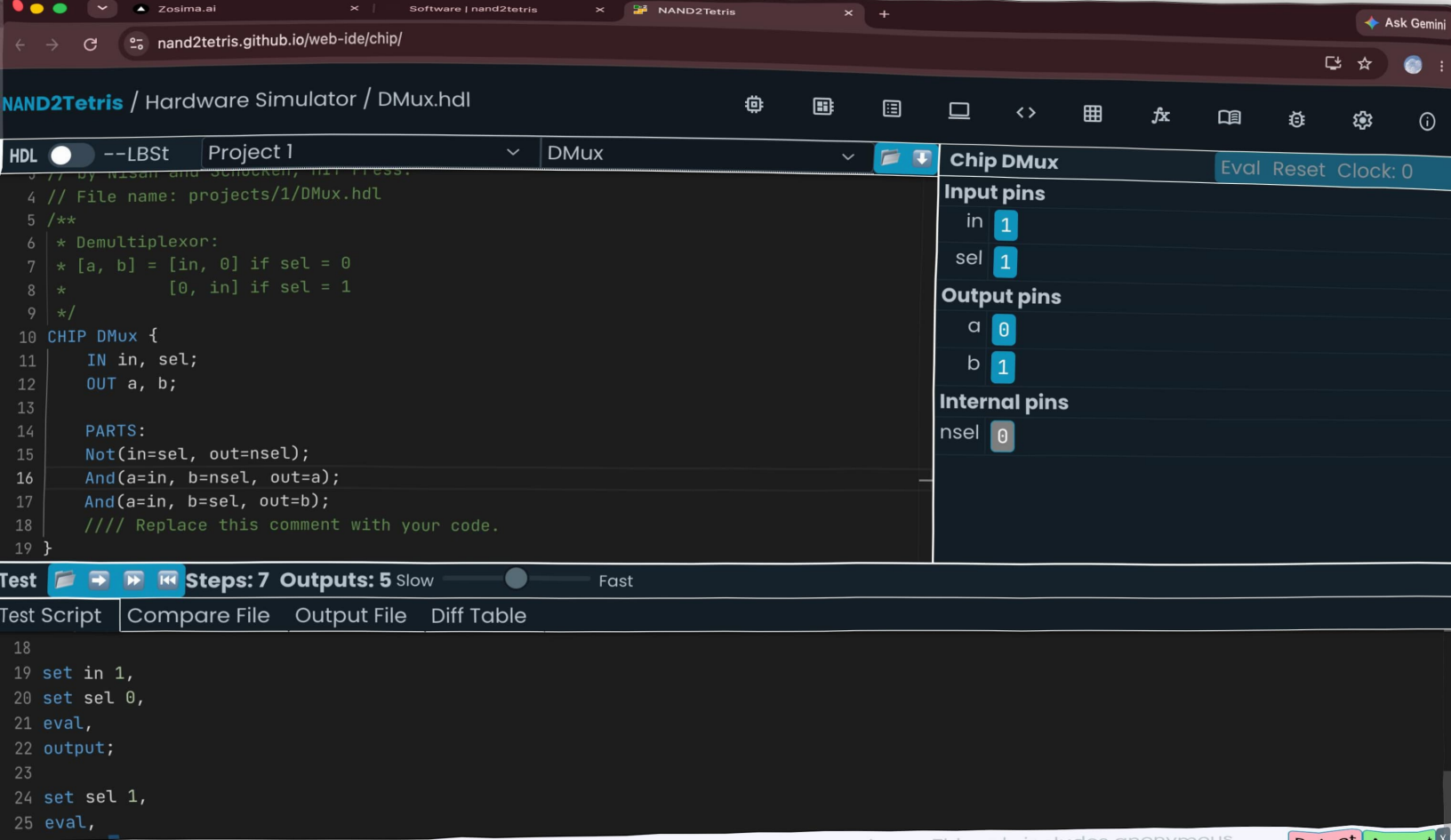
$$A \cdot \overline{B} + B \cdot \overline{A}$$

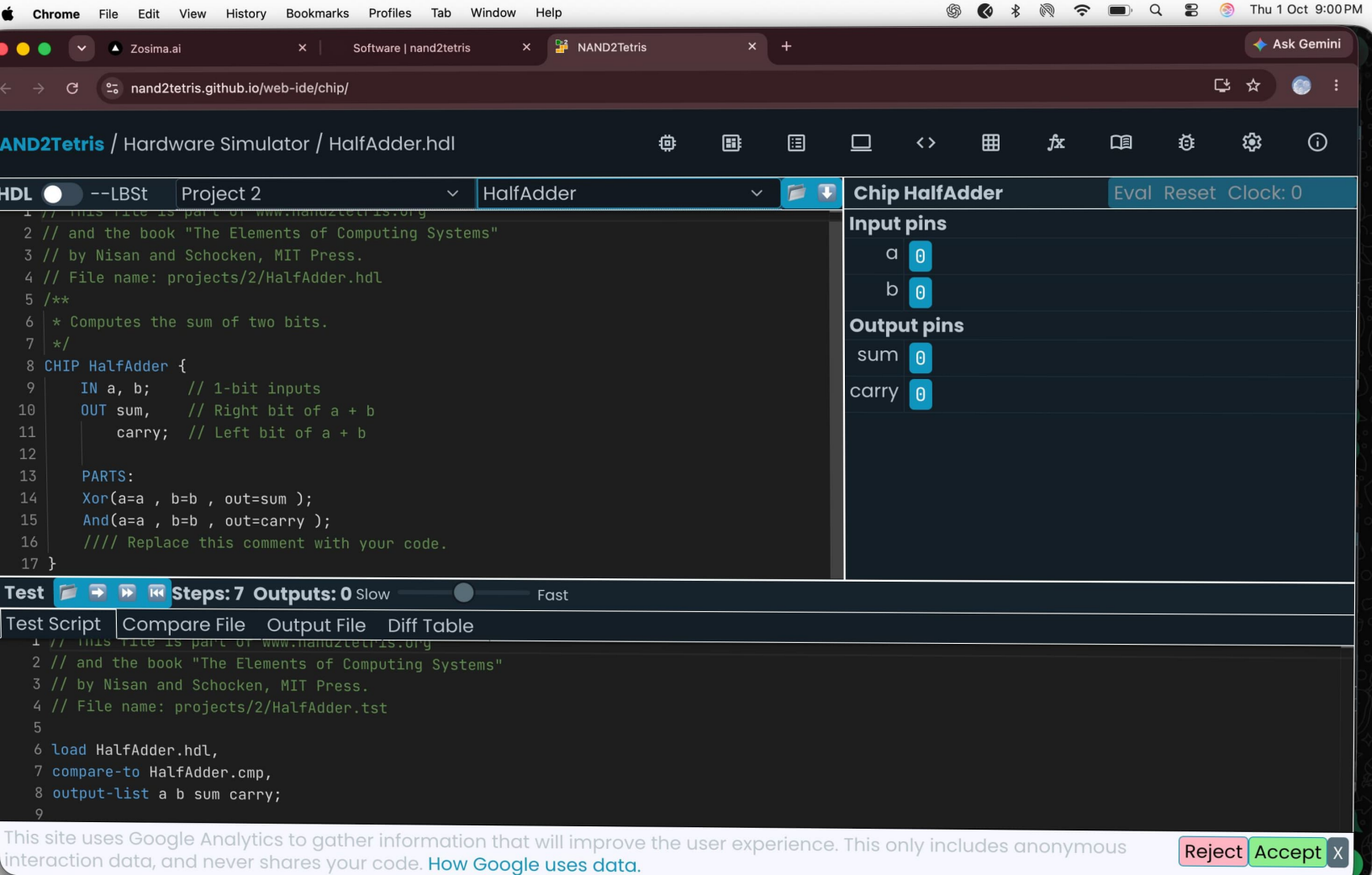

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NAND2Tetris / Hardware Simulator / And.hdl
HDL
--LBSt
Project 1
And
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/And.hdl
/**
* And gate:
* if (a and b) out = 1, else out = 0
*/
CHIP And {
IN a, b;
OUT out;
PARTS:
Nand(a=a, b=b , out=nandout);
}
Chip And
Eval
Reset
Clock: 0
Input pins
a 1
b 1
Output pins
out 0
Internal pins
nandout 0
Test
Steps: 7 Outputs: 5
Slow
Fast
Test Script
Compare File
Output File
Diff Table
set a 1,
set b 0,
eval,
output;
set a 1,
set b 1,
eval,
Reject
Accept

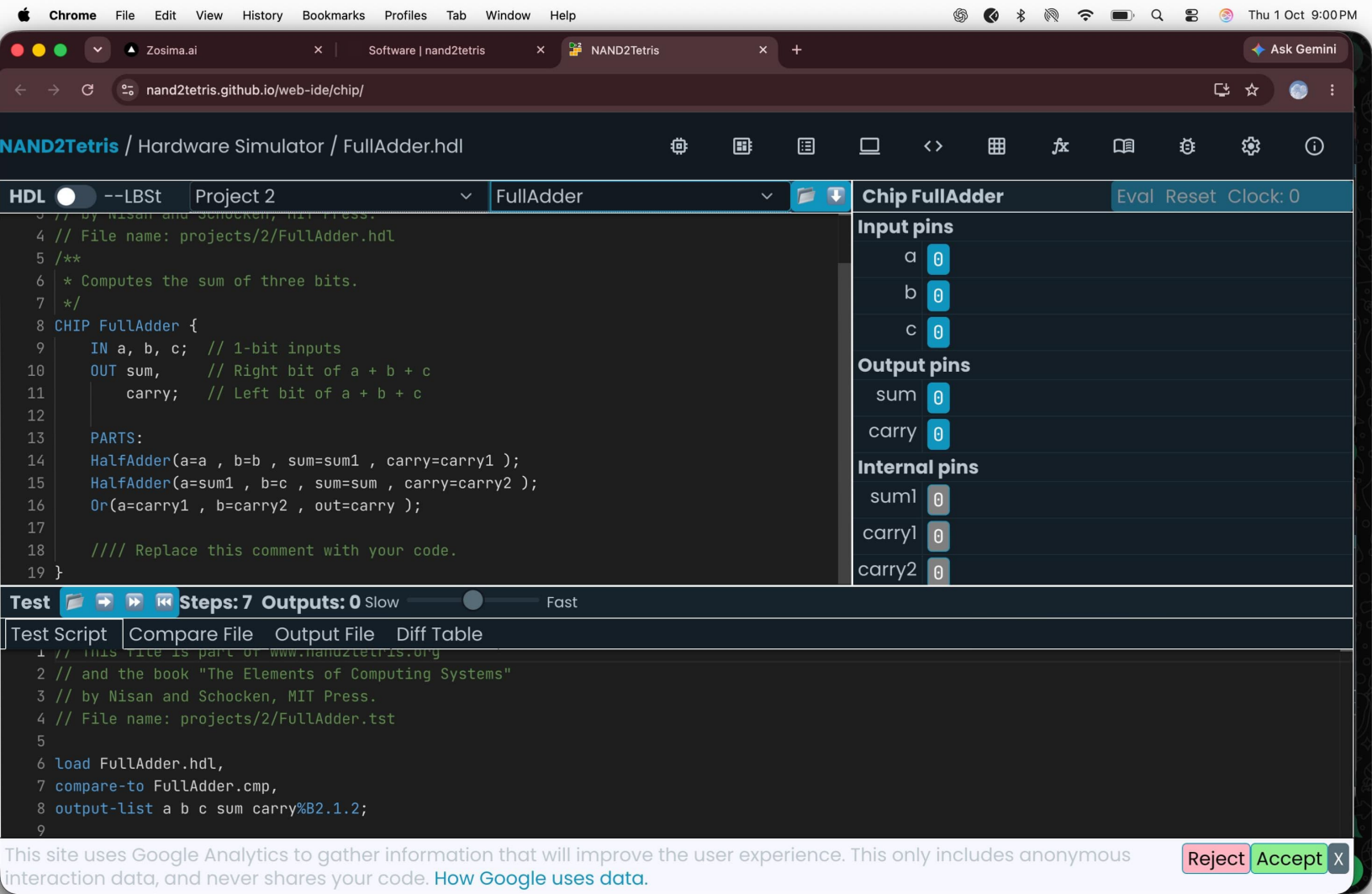

nand2tetris.github.io/web-ide/chip/
NAND2Tetris / Hardware Simulator / Or.hdl
HDL --LBSt
Project 1
Or
// File name: projects/1/Or.hdl
/**
* Or gate:
* if (a or b) out = 1, else out = 0
*/
CHIP Or {
IN a, b;
OUT out;
PARTS:
Nand(a=a, b=a, out=nota);
Nand(a=b, b=b, out=notb);
Nand(a=nota, b=notb, out=out);
//// Replace this comment with your code.
}
Chip Or
Eval Reset Clock: 0
Input pins
a 1
b 1
Output pins
out 1
Internal pins
nota 0
notb 0
Test
Steps: 7 Outputs: 5 Slow Fast
Test Script
Compare File
Output File
Diff Table
set a 1,
set b 0,
eval,
output;
set a 1,
set b 1,
eval,

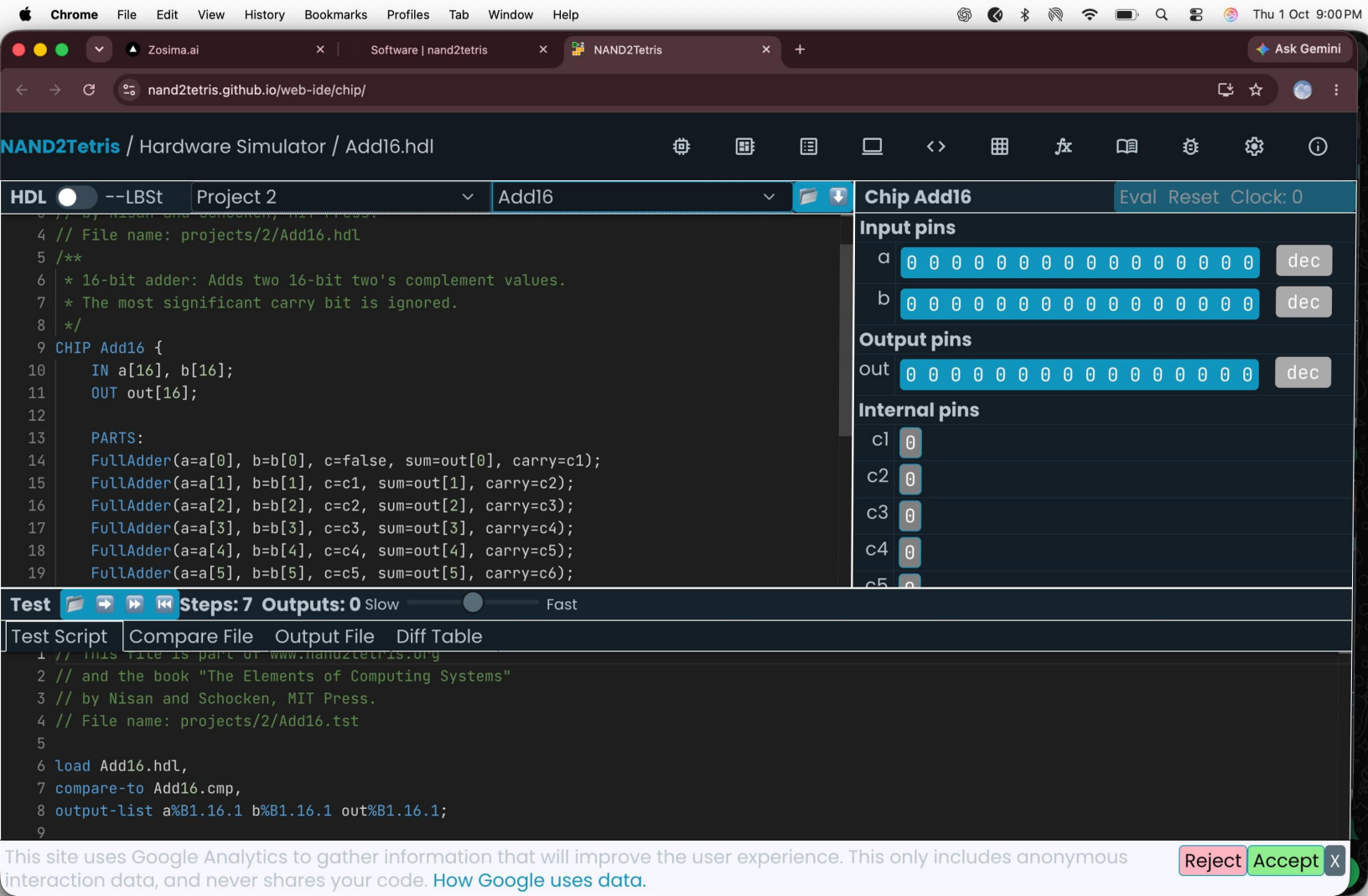

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NAND2Tetris / Hardware Simulator / Xor.hdl
HDL --LBSt Project 1 Xor
// File name: projects/1/Xor.hdl
/**
* Exclusive-or gate:
* if ((a and Not(b)) or (Not(a) and b)) out = 1, else out = 0
*/
CHIP Xor {
IN a, b;
OUT out;
PARTS:
Nand(a=a , b= b, out=n1);
Nand(a=a , b=b , out=n2);
Nand(a=b, b=n1 , out=n3);
Nand(a=n2 , b=n3 , out=out);
//// Replace this comment with your code.
}
Chip Xor Eval Reset Clock: 0
Input pins
a 0
b 0
Output pins
out 0
Internal pins
n1 1
n2 1
n3 1
Test Steps: 7 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/1/Xor.tst
load Xor.hdl,
compare-to Xor.cmp,
output-list a b out;
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Reject Accept X

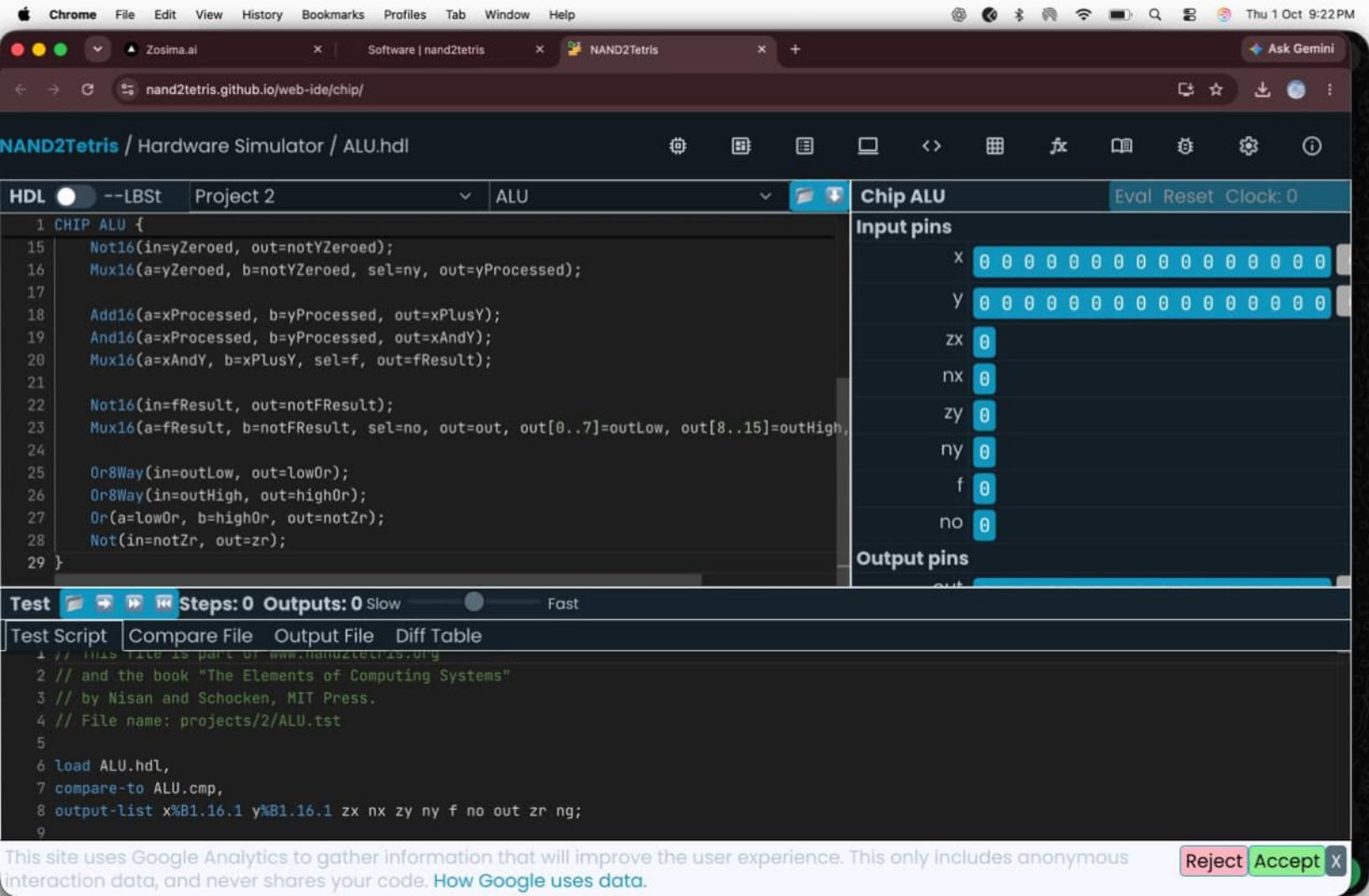

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NAND2Tetris / Hardware Simulator / Mux.hdl
HDL --LBSt
Project 1
Mux
1 // This file is part of www.nand2tetris.org
2 // and the book "The Elements of Computing Systems"
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Mux.hdl
5 /**
6 * Multiplexor:
7 * if (sel = 0) out = a, else out = b
8 */
9 CHIP Mux {
10 IN a, b, sel;
11 OUT out;
12
13 PARTS:
14 Not(in=sel, out=nsel);
15 And(a=a, b=nsel, out=outA);
16 And(a=b, b=sel, out=outB);
Chip Mux
Eval Reset Clock: 0
Input pins
a 1
b 1
sel 1
Output pins
out 1
Internal pins
nsel 0
outA 0
outB 1
Test
Steps: 11 Outputs: 9 Slow
Fast
Test Script
Compare File
Output File
Diff Table
40 set a 1,
41 set b 1,
42 set sel 0,
43 eval,
44 output;
45
46 set sel 1,
47 eval,
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NAND2Tetris / Hardware Simulator / DMux.hdl
HDL
--LBSt
Project 1
DMux
// File name: projects/1/DMux.hdl
/**
* Demultiplexor:
* [a, b] = [in, 0] if sel = 0
*   [0, in] if sel = 1
*/
CHIP DMux {
IN in, sel;
OUT a, b;
PARTS:
Not(in=sel, out=nsel);
And(a=in, b=nsel, out=a);
And(a=in, b=sel, out=b);
//// Replace this comment with your code.
}
Chip DMux
Eval
Reset
Clock: 0
Input pins
in 1
sel 1
Output pins
a 0
b 1
Internal pins
nsel 0
Test
Steps: 7 Outputs: 5
Slow
Fast
Test Script
Compare File
Output File
Diff Table
set in 1,
set sel 0,
eval,
output;
set sel 1,
eval,


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Software | nand2tetris
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nand2tetris.github.io/web-ide/chip/
AND2Tetris / Hardware Simulator / HalfAdder.hdl
HDL --LBSt
Project 2
HalfAdder
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/HalfAdder.hdl
/**
* Computes the sum of two bits.
*/
CHIP HalfAdder {
IN a, b; // 1-bit inputs
OUT sum, // Right bit of a + b
carry; // Left bit of a + b
PARTS:
Xor(a=a , b=b , out=sum);
And(a=a , b=b , out=carry);
//// Replace this comment with your code.
}
Chip HalfAdder
Eval Reset Clock: 0
Input pins
a 0
b 0
Output pins
sum 0
carry 0
Test
Steps: 7 Outputs: 0 Slow
Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/HalfAdder.tst
load HalfAdder.hdl,
compare-to HalfAdder.cmp,
output-list a b sum carry;
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NAND2Tetris / Hardware Simulator / FullAdder.hdl
HDL --LBSt
Project 2
FullAdder
// by Nisan and Schocken, MIT Press.
// File name: projects/2/FullAdder.hdl
/**
* Computes the sum of three bits.
*/
CHIP FullAdder {
IN a, b, c; // 1-bit inputs
OUT sum, // Right bit of a + b + c
carry; // Left bit of a + b + c
PARTS:
HalfAdder(a=a , b=b , sum=sum1 , carry=carry1);
HalfAdder(a=sum1 , b=c , sum=sum , carry=carry2);
Or(a=carry1 , b=carry2 , out=carry);
//// Replace this comment with your code.
}
Chip FullAdder
Eval Reset Clock: 0
Input pins
a 0
b 0
c 0
Output pins
sum 0
carry 0
Internal pins
sum1 0
carry1 0
carry2 0
Test
Steps: 7 Outputs: 0 Slow
Fast
Test Script
Compare File
Output File
Diff Table
// This file is part of www.nand2tetris.org
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/FullAdder.tst
load FullAdder.hdl,
compare-to FullAdder.cmp,
output-list a b c sum carry%B2.1.2;
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NAND2Tetris / Hardware Simulator / Add16.hdl
HDL --LBSt Project 2 Add16
// File name: projects/2/Add16.hdl
/**
* 16-bit adder: Adds two 16-bit two's complement values.
* The most significant carry bit is ignored.
*/
CHIP Add16 {
IN a[16], b[16];
OUT out[16];
PARTS:
FullAdder(a=a[0], b=b[0], c=false, sum=out[0], carry=c1);
FullAdder(a=a[1], b=b[1], c=c1, sum=out[1], carry=c2);
FullAdder(a=a[2], b=b[2], c=c2, sum=out[2], carry=c3);
FullAdder(a=a[3], b=b[3], c=c3, sum=out[3], carry=c4);
FullAdder(a=a[4], b=b[4], c=c4, sum=out[4], carry=c5);
FullAdder(a=a[5], b=b[5], c=c5, sum=out[5], carry=c6);
Chip Add16
Eval Reset Clock: 0
Input pins
a 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
b 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
Output pins
out 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 dec
Internal pins
c1 0
c2 0
c3 0
c4 0
Test Steps: 7 Outputs: 0 Slow Fast
Test Script Compare File Output File Diff Table
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/Add16.tst
load Add16.hdl,
compare-to Add16.cmp,
output-list a%B1.16.1 b%B1.16.1 out%B1.16.1;
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NAND2Tetris / Hardware Simulator / ALU.hdl
HDL --LBSt
Project 2
ALU
CHIP ALU {
Not16(in=yZeroed, out=notYZeroed);
Mux16(a=yZeroed, b=notYZeroed, sel=ny, out=yProcessed);
Add16(a=xProcessed, b=yProcessed, out=xPlusY);
And16(a=xProcessed, b=yProcessed, out=xAndY);
Mux16(a=xAndY, b=xPlusY, sel=f, out=fResult);
Not16(in=fResult, out=notFResult);
Mux16(a=fResult, b=notFResult, sel=no, out=out, out[0..7]=outLow, out[8..15]=outHigh,
Or8Way(in=outLow, out=lowOr);
Or8Way(in=outHigh, out=highOr);
Or(a=lowOr, b=highOr, out=notZr);
Not(in=notZr, out=zr);
}
Chip ALU
Eval Reset Clock: 0
Input pins
x 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
y 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
zx 0
nx 0
zy 0
ny 0
f 0
no 0
Output pins
Test
Steps: 0 Outputs: 0 Slow
Fast
Test Script
Compare File
Output File
Diff Table
// and the book "The Elements of Computing Systems"
// by Nisan and Schocken, MIT Press.
// File name: projects/2/ALU.tst
load ALU.hdl,
compare-to ALU.cmp,
output-list x%B1.16.1 y%B1.16.1 zx nx zy ny f no out zr ng;
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Reject
Accept
X

ALU Function table

zx	nx	zy	ny	f	no	Function
1	0	1	0	1	0	0
1	1	1	1	1	1	1
1	1	1	0	1	0	−1
0	0	1	1	0	0	x
1	1	0	0	0	0	y
0	0	1	1	0	1	!x
1	1	0	0	0	1	!y
0	1	1	1	1	1	−x
1	1	0	1	1	1	−y
0	1	1	1	1	0	x+1
1	1	0	1	1	0	y+1
0	0	1	1	1	0	x−1
1	1	0	0	1	0	y−1
0	0	0	0	1	0	x+y
0	1	0	0	1	1	x−y
0	0	0	1	1	1	y−x
0	0	0	0	0	0	x&y
0	1	0	1	0	1	x|y